

TWIN CITIES ORTHOPEDICS ACL KNEE STUDY



**PAIRED SAMPLE
T-TEST
ANALYSIS AND
TIME SERIES
ANALYSIS:**

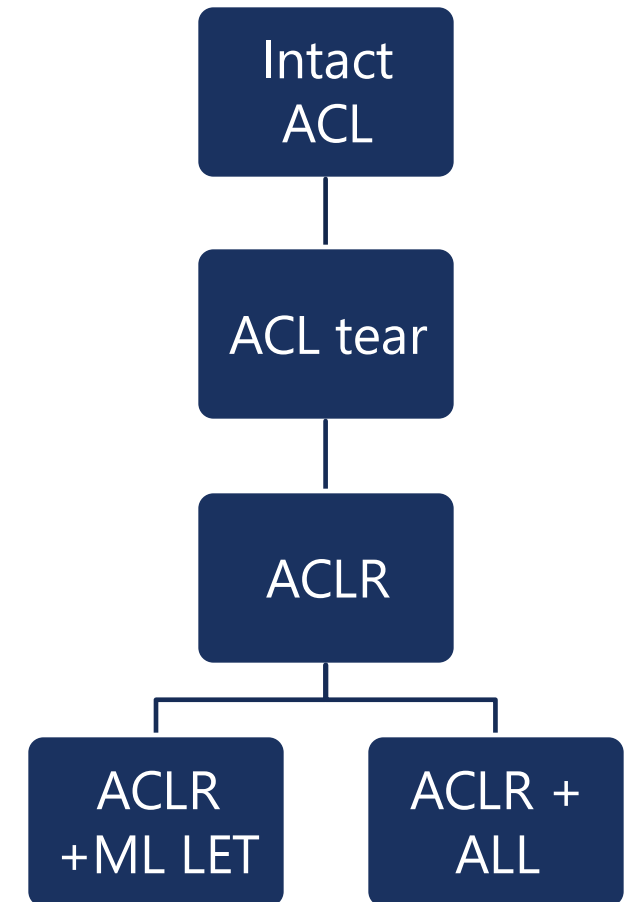
**JACKSON
SPORER**

POPULATION OF INTEREST

- Sample of 20 cadaveric knees
- Interest in individuals needing ACL reconstruction surgery
- Evaluation of alternative reconstruction methods that may yield better outcomes than current standard
- Numerous ACL reconstructions experience re-rupture
 - Aiming to identify methods that could reduce this risk
- Could improve ACL surgical standards and provide longer lasting results

DATA COLLECTION

- 20 cadaveric knees
 - 10 left, 10 right
- Each undergoes an ACL tear
- Followed by a standard ACL reconstruction (ACLR)
- Then split into two groups
 - Modified Lemaire Lateral Extra-Articular Tenodesis (ML LET)
 - Anatomic Anterolateral Ligament (ALL)



DATA COLLECTION

- Under each surgical condition, mechanical tests are performed by a machine

Internal Rotation

Twisting force

5 Nm torque

0-90° flexion at 15° increments

Anterior Tibial Load

Stretches ACL

88N anterior force

30° and 90° flexion

Simulated Pivot Shift

5 Nm internal rotation technique

10 Nm valgus torque

15° and 30° flexion



DATA COLLECTION

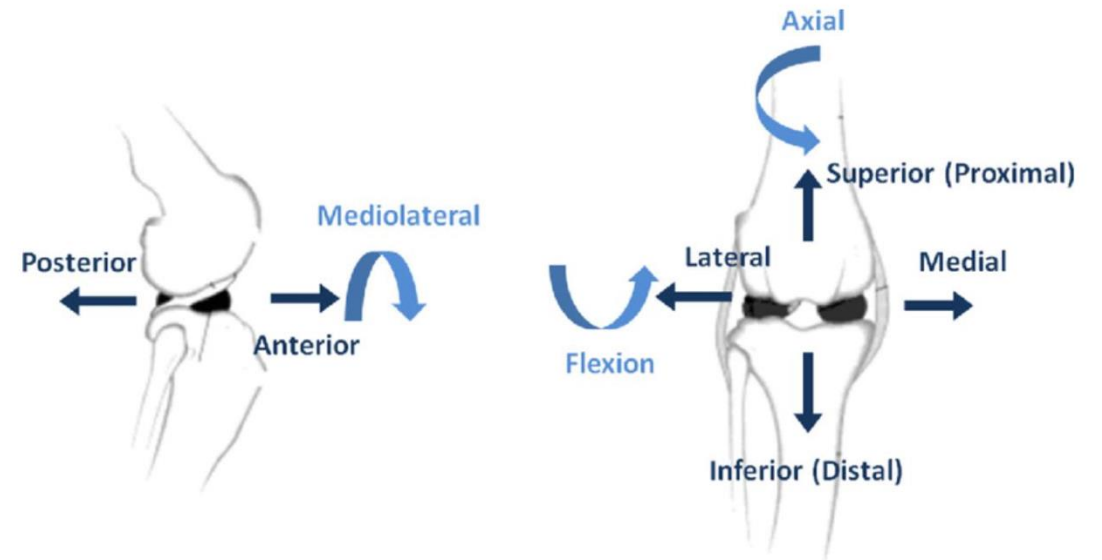
- The mechanical tests measure three directions and three rotations

Directions

- Medial
- Posterior
- Superior

Rotations

- Flexion
- Valgus
- Internal Rotation



PROBLEM AND GOALS

- ML LET reduces internal rotation in an ACL deficient knee, but ACLR is still required to restore anterior tibial translation
- ML LETs used to be a more common addition to ACLR, but fell out of favor due to increased risk of lateral compartment over-constraint and graft elongation
- Does an isolated ACL reconstruction restore knee stability?
 - If not, does either ML LET or ALL help?
- Are there mechanical differences between ML LET/ALL?

DISCUSS DATA CLEANSING AND MANIPULATION

- Removal of invalid data and transformations are considered and performed where appropriate
- Cleaned Data by adjusting values. Subtracted Intact by Equilibrium to get new Intact. Then subtracted everything else by the raw Intact.
 - Clean, long form in excel
 - 20 specimen recorded (20 knees, L and R from each specimen)
 - There are 4 surgical states (Intact, ACLR, ML LET, ALL)
 - 11 different tests ran on each specimen
 - Each row of data represents 1 millisecond of a test being ran
 - The degree of each knee movement is recorded at each millisecond the test is ran for



PAIRED SAMPLE T-TEST ANALYSIS



VARIABLES

- Our first variable was **State** which includes the different surgeries. ACL Intact, ACL Deficient, ACLR, ACLR + ALL, and ACLR + ML LET.
- Our second variable was **Trajectory**. Which includes IR, Pivot, and ATT.
- Within each trajectory there are different flexions/angles in which the knee is measured at.
- Other variables include the different knee rotations. Which includes Medial, Posterior, Superior, Valgus, and IR. These different knee rotations allowed us to simulate real world scenarios for knee stability.

IDENTIFY RESEARCH GOALS AND HYPOTHESES

- Finding out how the state of the knee impacts the ability of the knee to rotate when using different trajectories
- Looking at which surgery type led to results that were most similar to the intact state of the knee
- Understand the limitations and benefits of ALL and ML LET surgeries
- Our hypothesis is that the ACLR + ALL surgery will perform the best overall.

WHY NOT ANOVA?

Measured Values:

- For our analysis intact knee used raw values
- Every other state used measured differences from the Intact State
- Thus, we couldn't rely on our ANOVA analysis as it relies on comparing raw values against each other.
- Therefore, we ran 44 paired sample t-tests for each of the 5 rotations to compare every other state against the intact state

APPLY ANALYTICAL METHODS TO PRESENT DATA

- $T\text{-score} = \text{mean difference} / \sqrt{(\text{standard deviation}^2 / \text{sample size})}$
- Mean difference = Averaged measured differences from State – Raw Intact at the 5 different knee rotations for every unique combination of trajectory and State
- Standard Deviation: Standard Deviation for every State being compared to the Intact Knee
- Compared this T-Score to a theoretical T-distribution using 9 degrees of freedom for ACLR + ALL and ACLR + ML LET states and 19 degrees of freedom for all other states
- Used a Bonferroni Correction for our paired sample t-tests due to sufficient amount of t-tests
- Our alpha level was 0.001136 calculated by dividing a normal alpha level, 0.05, by the number of groups we had, 44.

PRESENT AND INTERPRET COMPUTER MODEL OUTPUT

Posterior

Trajectory	Intact (n = 20)	ACLR (n = 20)	ACLR + ALL (n = 10)	ACLR + ML LET (n = 10)	ACLR Deficient (n = 20)
ATT 30	-5.44 ± 2.81	-3.11 ± 3.63 ⁱ	-5.44 ± 3.22 ⁱ	-3.86 ± 4.36	-7 ± 3.9 ⁱ
ATT 90	-4.43 ± 1.80	-2.51 ± 2.07 ⁱ	-3.98 ± 3.43	-2.17 ± 2.71	-4.4 ± 2.9 ⁱ
IR 0	0.11 ± 1.83	-0.92 ± 1.20	-1.59 ± 1.25	0.29 ± 1.61	-1.9 ± 1.2 ⁱ
IR 15	0.18 ± 1.62	-0.97 ± 0.86 ⁱ	-1.67 ± 1.32	0.79 ± 0.96	-1.4 ± 0.9 ⁱ
IR 30	0.61 ± 1.27	-0.87 ± 0.74 ⁱ	-1.82 ± 1.15 ⁱ	0.61 ± 0.79	-0.8 ± 0.7 ⁱ
IR 45	2.28 ± 2.73	-0.51 ± 0.85	-1.97 ± 1.35	0.54 ± 1.01	-0.4 ± 0.8
IR 60	1.03 ± 1.33	-0.29 ± 0.87	-1.45 ± 0.96 ⁱ	0.37 ± 1.07	-0.2 ± 0.8
IR 75	1.09 ± 1.37	-0.19 ± 0.79	-1.06 ± 0.82	0.34 ± 0.98	-0.1 ± 0.7
IR 90	1.14 ± 1.38	-0.15 ± 0.65	1.37 ± 6.76	0.31 ± 0.78	-0.1 ± 0.6
Pivot 15	0.13 ± 2.07	-1.13 ± 1.18 ⁱ	-2.05 ± 1.12 ⁱ	0.79 ± 1.16	-1.8 ± 1.2 ⁱ
Pivot 30	0.52 ± 1.65	-0.95 ± 1.25	-2.13 ± 1.09 ⁱ	0.48 ± 1.41	-1.2 ± 1.3 ⁱ

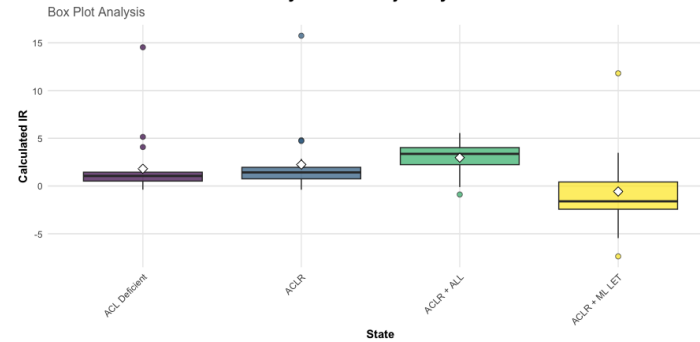
IR

Trajectory	Intact (n = 20)	ACLR (n = 20)	ACLR + ALL (n = 10)	ACLR + ML LET (n = 10)	ACL Deficient (n = 20)
ATT 30	-2.0 ± 4.5	0.4 ± 4.4	1.1 ± 3.1	-2.9 ± 5.8	-1.6 ± 4.9
ATT 90	-1.6 ± 2.3	-0.7 ± 2.8	-1.6 ± 3.5	-3.9 ± 5.0	-2.2 ± 3.5
IR 0	13.3 ± 5.5	2.3 ± 3.1	1.4 ± 2.5	-0.5 ± 5.1	2.1 ± 3.0
IR 15	10.9 ± 4.0	2.3 ± 3.4	1.6 ± 1.7	-0.8 ± 5.1	1.9 ± 3.2
IR 30	10.5 ± 3.2	2.2 ± 3.3	3.0 ± 2.0 ⁱ	-0.6 ± 5.1	1.8 ± 3.1
IR 45	12.5 ± 4.0	2.0 ± 2.8	3.5 ± 2.9	-0.4 ± 4.9	1.6 ± 2.7
IR 60	8.7 ± 2.7	1.9 ± 2.6	3.4 ± 2.7	-0.6 ± 4.9	1.5 ± 2.5
IR 75	8.1 ± 2.4	1.8 ± 2.7	2.4 ± 2.9	-1.1 ± 5.4	1.5 ± 2.7
IR 90	8.1 ± 2.3	1.8 ± 3.0	1.3 ± 3.5	-1.9 ± 6.3	1.5 ± 3.0
Pivot 15	12.5 ± 5.8	2.9 ± 5.8	2.1 ± 1.9	-0.1 ± 8.4	2.5 ± 5.7
Pivot 30	11.8 ± 4.8	2.9 ± 6.3	3.1 ± 1.9 ⁱ	0.2 ± 9.2	2.5 ± 6.2

* i denotes statistically significantly different from Intact. *Intact column displays the raw value for Intact and the rest of the columns show the respective differences from the Intact state

GRAPHICS TO SHOW DIFFERENCES FROM INTACT

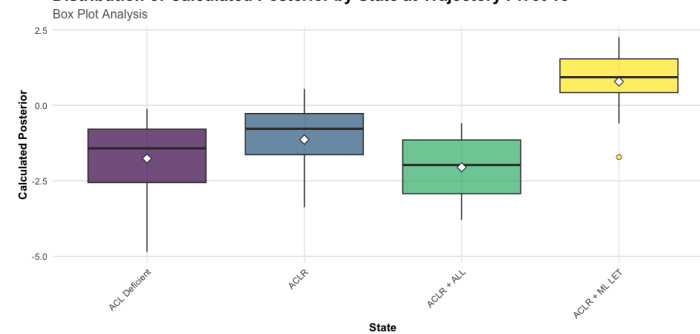
Distribution of Calculated IR by State at Trajectory IR 30



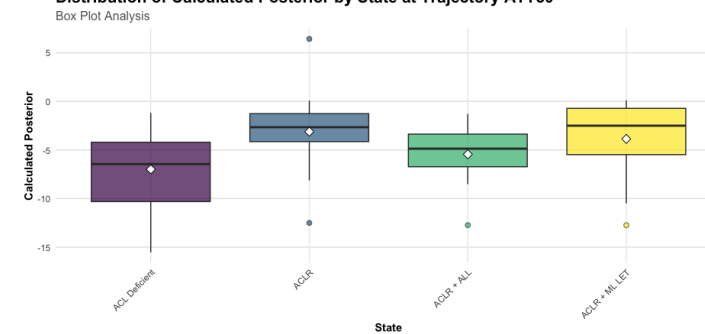
Distribution of Calculated Posterior by State at Trajectory IR 30



Distribution of Calculated Posterior by State at Trajectory Pivot 15



Distribution of Calculated Posterior by State at Trajectory ATT30



DESCRIBE PRACTICAL SIGNIFICANCE OF RESULTS

- 17 significant differences from Intact State for the Posterior rotation and only 2 significant differences for the IR rotation.
- These two rotations above were considered the "rotations" of focus
- Also, the other 3 rotations: Valgus, Medial, or Superior only had 1 significant value between these 3 rotations
- Based on our results, ACLR + LET is the most similar ACL reconstruction surgery to the Intact Knee.
- However, ACLR + ML LET restricts the knees mobility compared to the intact knee.

RESULTS IN CONTEXT

- TCO and other companies focused on ACL reconstruction can use our results to further understand the benefits and limitations of each surgery
- For example, if a patient wants more stability in their knee, a doctor may recommend ACLR + ML LET reconstruction surgery for the ACL.
- However, if mobility of the knee is of concern, then ACLR + ALL might be the best surgery
- Furthermore, doctors can also use our research to understand how the knee performs under different surgeries with different knee movements

EXPLAIN MODEL LIMITATIONS AND VIOLATIONS

Model Limitations:

- Limited number of Specimens (20 specimens)
- For ACLR + ML LET and ACLR + ALL has a sample size of 10
- Due to the limited sample size and the limitations that come with it, we could repeat this study or process in the future with a larger sample size to see if our results are similar or much different.



ARIMA ANALYSIS



INFERENCE MODEL SELECTION

- Autoregressive Integrated Moving Average (ARIMA)
- Technique for forecasting possible future values of a time series
- Evaluate how knee movements progressed over time
- How closely reconstructed knees matched behavior of intact knees

VARIABLE SELECTION

- Data from four knees
 - Two left, two right
 - Evenly divided between ML LET and ALL groups
- Pivot 30° flexion data only
- Four key movements
 - Medial, posterior, valgus, internal rotation
- Every millisecond, the machine recorded the angle of the knees

RESEARCH GOALS AND HYPOTHESES

Goal 1: Determine which reconstruction method returned the knee closest to its intact, pre-injury state

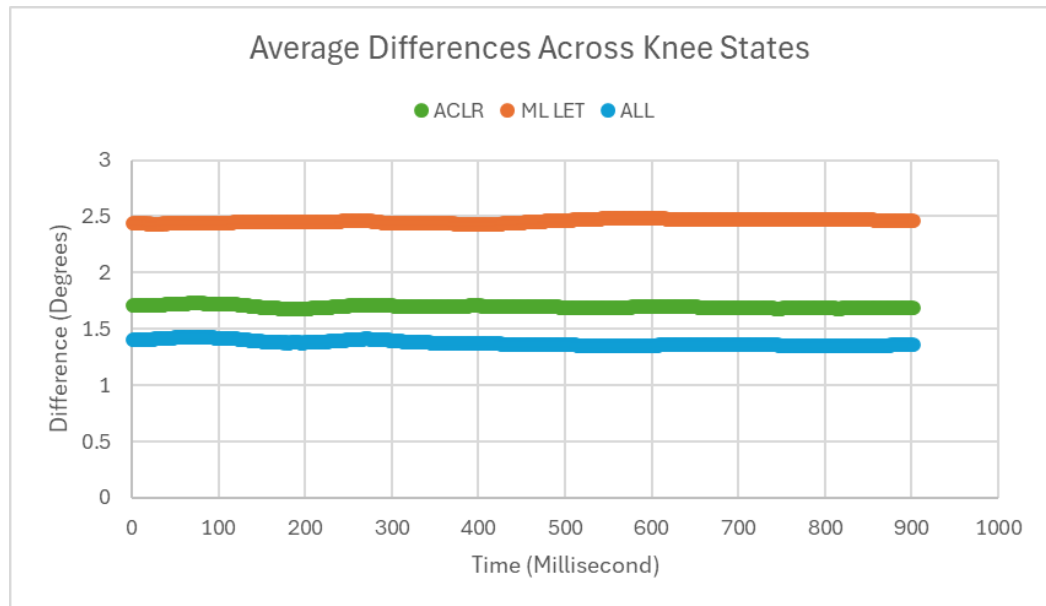
Hypothesis: ML LET would produce results closest to its intact state

- Based on historical popularity in the medical field

Goal 2: Create ARIMA forecasts of knee movements over time

Hypothesis: ML LET would achieve stationary closest to zero

SUMMARY STATISTICS



- Average difference in degrees between each reconstructed state and the intact state across all samples

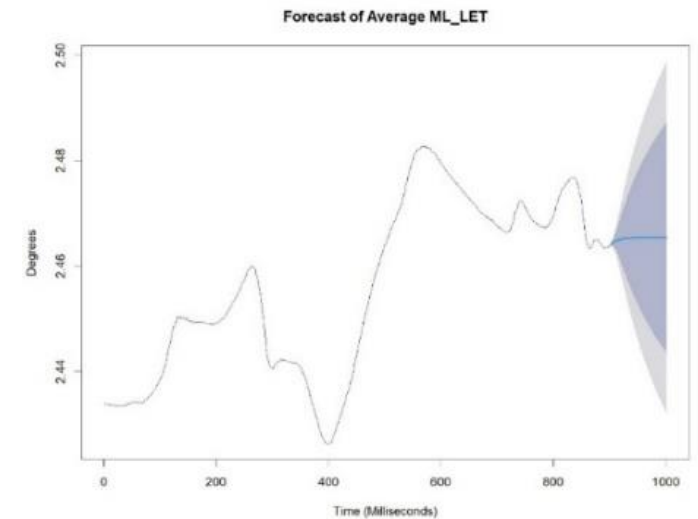
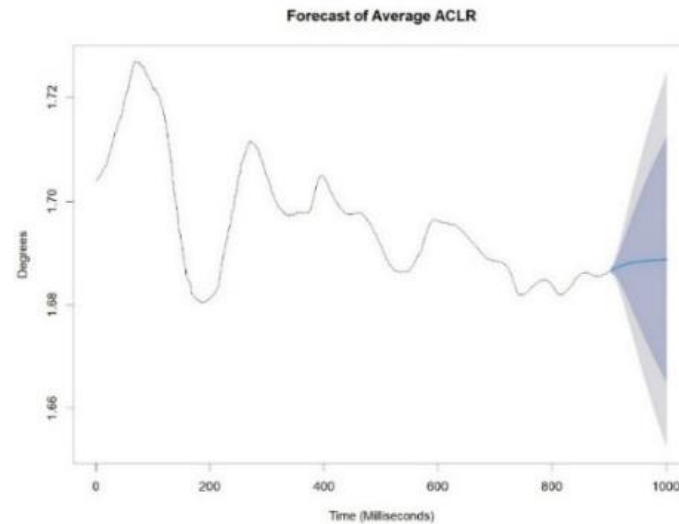
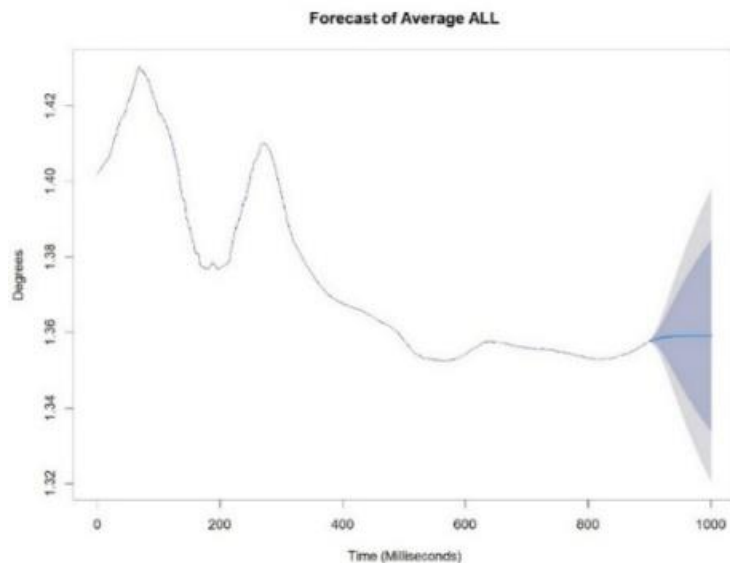
1. ALL $\approx 1.4^\circ$
2. ACLR $\approx 1.75^\circ$
3. ML LET $\approx 2.5^\circ$

APPLICATION OF STATISTICAL MODEL

- ARIMA model applied to each reconstruction method
- Each model was trained on the first 903 milliseconds of data
 - Used to predict the following 100 milliseconds
- Evaluate consistency and stability of each method over time

COMPUTER OUTPUT

- Forecasts estimate expected path over the next 100 milliseconds
- 90% and 95% confidence intervals provided in the shaded area
- ACLR and ML LET plots have wider forecast ranges ALL shows narrower forecast



STATISTICAL SIGNIFICANCE

- To test each ACL reconstruction, we used the Augmented Dickey-Fuller test (ADF) and KPSS test
- ML LET method failed to show significant evidence of stationarity
 - $p = 0.3135 > 0.05$
- ACLR and ALL showed statistically significant evidence of stationarity
 - $p = 0.01 < 0.05$
- KPSS further proved ML LET's non-stationary status
 - $p = 0.01 < 0.05$

```
> adf_test_results[["Average ML_LET"]]

Augmented Dickey-Fuller Test

data: ts_data
Dickey-Fuller = -2.6256, Lag order = 9,
p-value = 0.3135
alternative hypothesis: stationary
```

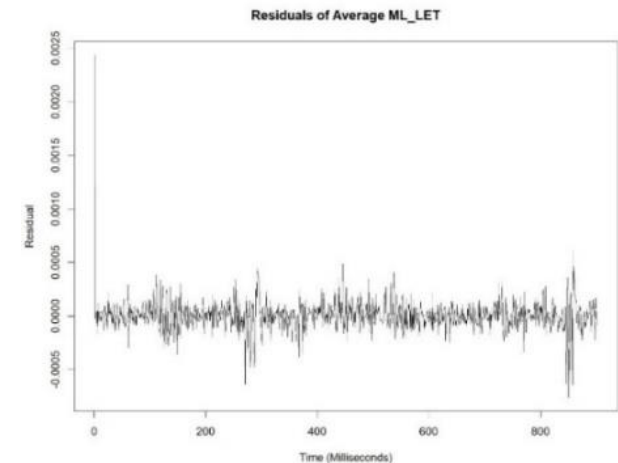
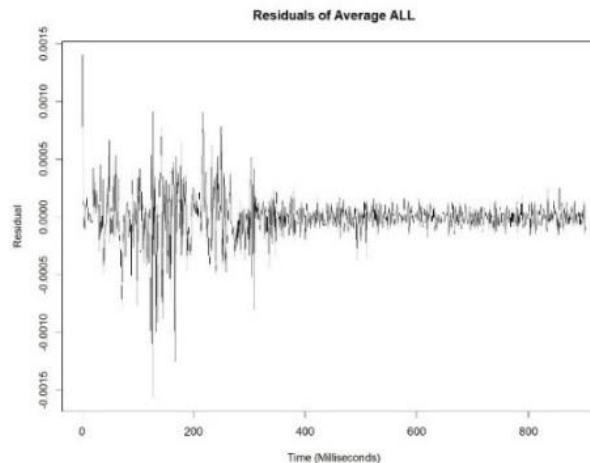
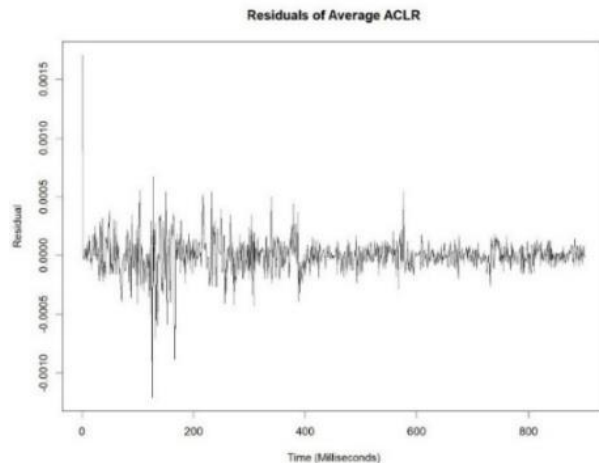
```
> kpss_test_results[["Average ML_LET"]]

KPSS Test for Level Stationarity

data: ts_data
KPSS Level = 7.9946, Truncation lag parameter = 6, p-value = 0.01
```

MODEL CONDITIONS

- Residual plots support the results of the ADF test
- Goal is to see variation in residuals early on, followed by stabilization around zero as the plot progresses
- ACLR and ALL show expected, statistically significant pattern
- ML LET shows non-stationary results
 - Minimum early variation, increased fluctuation later in the plot



METHODOLOGY EVALUATION

- Like all research, this study comes with limitations
- Research only consists of data from four knees
 - Difficult to draw broad conclusions
- Key insights not included in original data
 - Activity level, weight, age, etc.
- Despite constraints, the study offers valuable insights while recognizing the need for further research



CONCLUSIONS



FUTURE RESEARCH

Paired Sample T-Test

- Exploring the specific interaction between the state and trajectory interaction could be beneficial.
- We could also explore more advanced statistics like ANCOVA to compare state means with a covariate.
- MANCOVA could be done to allow us to see if state and trajectory have a significant effect on the different rotations.

ARIMA

- Explore knees from different flexion points
- Same study, larger sample
- Collect data over longer time periods, to observe how each reconstruction method behaves during recovery

The background of the slide is a dense, overlapping collection of numerous small, rectangular, light-colored cards. Each card features a large, bold, black question mark. The cards are scattered across the entire frame, creating a textured, busy appearance. On the left side, there is a solid dark blue vertical rectangle that serves as a backdrop for the text.

THANK YOU!

QUESTIONS?